Numerical Solutions to the One-dimensional Schrödinger's Equation

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Abstract

Throughout this project, four different numerical methods were used in order to obtain solutions to the one-dimensional Schrödinger equation, with a Gaussian wavepacket as initial condition. The methods implemented were Forward and Backward Euler, Crank-Nicolson and Split Step Fourier. The time evolution of this wavepacket was first studied in three potentials with familiar analytic solutions, which provided the possibility of clearly evaluating the level of precision of each of the schemes considered. Furthermore, comparing the performances of these four methods enabled us to determine the most efficient one among them, which was then used in order to study four other potentials, for which the solutions are essential in several areas of physics, but extremely hard or rather impossible to determine analytically.

1 Introduction

Despite the highly abstract level of quantum mechanics that sometimes raises more questions than it answers, one undeniable fact is that the time evolution of any quantum state is governed by the Schrödinger Equation. Finding itself at the very core of Quantum Mechanics, solving this equation has proved to be both unavoidable and also surprisingly challenging over the years. As it is a partial differential equation, attempting to solve it has yielded either particularly complicated solutions for the cases of certain potentials, or no analytic solutions at all.

The paramount importance of solving this equation has naturally brought about the idea of using numerical methods in order to generate solutions for both the linear and non-linear cases, these methods proving to be extremely precise in producing relevant results for physical applications.

2 Theory

In this section, we will discuss 4 numerical methods used for solving the Schrödinger Equation (1) for a particle of mass m, in one dimension, for various potentials. In mathematical terms, what we need to do is solve the following initial value problem:

$$i\hbar\frac{\partial}{\partial t}\Psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\Psi(x,t) + V(x,t)\Psi(x,t)\,, \quad \text{with known} \quad \Psi(x,0). \tag{1}$$

2.1 Forward Euler method

Forward Euler is the easiest explicit method of numerical differentiation used for solving initial value problems. The space and time domains are treated as discrete, namely $x_n = x_0 + n\Delta x$ and $t_n = t_0 + n\Delta t$, where Δx is the step size length and Δt is the time step. We define $\alpha = \Delta t/\Delta x^2$. For simplicity, from here on we will refer to the position x_i as position i and to time t_j as time j. By using the standard approximation (forward formula) for derivatives with local approximation errors $O(\Delta t)$ and $O(\Delta x^2)$, after i steps and j time steps, we can write:

$$\frac{\partial\Psi}{\partial t} = \frac{\Psi(x,t+\Delta t)-\Psi(x,t)}{\Delta t} = \frac{\Psi(x_i,t_j+\Delta t)-\Psi(x_i,t_j)}{\Delta t} \approx \frac{\Psi_{i,j+1}-\Psi_{i,j}}{\Delta t}, \tag{2}$$

$$\frac{\partial^2\Psi}{\partial^2 x} \approx \frac{\Psi(x_i+\Delta x,t_j)-2\Psi(x_i,t_j)+\Psi(x_i-\Delta x,t_j)}{\Delta x^2} \approx \frac{\Psi_{i+1,j}-2\Psi_{i,j}+\Psi_{i-1,j}}{\Delta x^2}. \tag{3}$$

Substituting equations (2) and (3) into equation (1), we can construct the value of the wavefunction at position i and time $j+1$ from the values at time j and positions $i-1$, i and $i+1$.

In matrix form, this is written as $\Psi_{j+1} = (\hat{A}+\hat{V})\,\Psi_j = (I - \frac{i\hbar}{2m}\alpha\hat{B} + \hat{V})\,\Psi_j$, where:

$$\Psi_j = \begin{pmatrix} \Psi_{1,j} \\ \Psi_{2,j} \\ \cdots \\ \Psi_{n,j} \end{pmatrix} \quad \hat{V} = \begin{pmatrix} V(x) & 0 & \cdots & 0 \\ 0 & V(x) & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & V(x) \end{pmatrix} \quad \hat{B} = \begin{pmatrix} 2 & -1 & \cdots & 0 \\ -1 & 2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & 2 \end{pmatrix}, \tag{4}$$

noting that $\hat{B}$ is a tridiagonal matrix.

By the same reasoning, we can rewrite equation (1) as a set of matrix-vector multiplications:

$$\Psi_{j+1} = (\hat{A} + \hat{V})\,\Psi_j = (\hat{A} + \hat{V})^2\,\Psi_{j-1} = \cdots = (\hat{A} + \hat{V})^{j+1}\,\Psi_0.$$

Although it is fairly easy to compute, this method has a very weak stability condition [1]:

$$\alpha = \Delta t/\Delta x^2 \leq 1/2. \tag{5}$$

2.2 Backward Euler method

For the Backward Euler method, an implicit scheme will be used instead of the explicit one seen before, starting with the backward formula for the time differentiation:

$$\frac{\partial \Psi}{\partial t} = \frac{\Psi(x_i, t_j) - \Psi(x_i, t_j - \Delta t)}{\Delta t} \approx \frac{\Psi_{i,j} - \Psi_{i,j-1}}{\Delta t}. \tag{6}$$

For the spatial derivative, the formula previously given in equation (3) will be used.

When substituting equations (3) and (6) back into equation (1), it can be seen that the term Ψ_{j+1} appears on both sides of the equation, meaning that for this method we need to solve an algebraic equation for the unknown term, which reduces to inverting a matrix. This makes the backward Euler method stable, regardless of the sizes of the steps that are chosen [1].

The matrix $\hat{A}$ now becomes $\hat{A} = I + \frac{i\hbar}{2m}\alpha\hat{B}$, leading to the following recurrence relationship:

$$\Psi_j = (\hat{A} + \hat{V})^{-1}\,\Psi_{j-1} = \cdots = (\hat{A} + \hat{V})^{-j}\,\Psi_0.$$

2.3 Crank-Nicolson method

The implicit and explicit methods used for solving the Schrödinger Equation can be combined, by introducing a parameter θ [1]:

$$\begin{aligned} \frac{\theta\left(\Psi_{i-1,j} - 2\Psi_{i,j} + \Psi_{i+1,j}\right)}{\Delta x^2} + \frac{(1-\theta)\left(\Psi_{i+1,j-1} - 2\Psi_{i,j-1} + \Psi_{i-1,j-1}\right)}{\Delta x^2} = \\ = -\frac{2mi}{\hbar}\frac{\Psi_{i,j} - \Psi_{i,j-1}}{\Delta t} + \frac{2m}{\hbar^2}V(x)\Psi_{i,j}. \end{aligned} \tag{7}$$

In equation (7), the forward formula is recovered for $\theta = 0$ and for $\theta = 1$, the backward one. For the case $\theta = 1/2$, we obtain a new scheme called Crank-Nicolson, which yields a truncation in

time that goes like $O(\Delta t^2)$ and is stable regardless of the sizes of Δx and Δt [1].

Expressing Ψ_j in terms of the values of Ψ_{j-1}, we can write the Crank-Nicolson scheme as:

$$-\alpha\Psi_{i-1,j}+\Big(2\alpha-i\frac{4m}{\hbar}+\frac{4m}{\hbar^2}\Delta t\,V(x)\Big)\Psi_{i,j}-\alpha\Psi_{i+1,j}=\alpha\Psi_{i-1,j-1}+\Big(-2\alpha-i\frac{4m}{\hbar}\Big)\Psi_{i,j-1}+\alpha\Psi_{i+1,j-1}.$$

This can also be written in a matrix-vector form, using the matrices given in (4):

$$\Psi_j=\Big(\big(-i\frac{4m}{\hbar}\big)\hat{I}+\frac{4m}{\hbar^2}\Delta t\,\hat{V}+\alpha\hat{B}\Big)^{-1}\Big(\Big(-i\frac{4m}{\hbar}\Big)\hat{I}-\alpha\hat{B}\Big)\Psi_{j-1},$$

which enables us to determine any Ψ_j, provided that we know the initial state, Ψ_0.

2.4 Split Step Fourier method

The Split Step Fourier method for solving the one-dimensional Schrödinger equation implies splitting the Hamiltonian operator acting on the wavefunction in two parts and treating them separately, with an error of the order of the timestep squared [2]. Equation (1) can be more conveniently written in the following form:

$$\frac{\partial\Psi}{\partial t}=i\mathcal{T}\Psi-iV\Psi,\quad\text{where}\quad\mathcal{T}=\frac{\hbar}{2m}\frac{\partial^2}{\partial x^2}\quad\text{and}\quad V=\frac{V(x)}{\hbar}.\tag{8}$$

As a first instance, for a small time step Δt, we ignore the effect of $\mathcal{T}$ and we solve the straightforward x-space equation, obtaining the following evolution:

$$\Psi_V(x,t+\Delta t)=\Psi(x,t)e^{-iV(x)\Delta t/\hbar}.\tag{9}$$

Next, we evaluate the effect of $\mathcal{T}$ in the Fourier domain, which will reduce equation (8) to a straightforward k-space equation, giving the following evolution for a small Δt:

$$\widetilde{\Psi}(k,t+\Delta t)=\widetilde{\Psi}_V(k,t)e^{-i\mathcal{F}(T)\Delta t}=\widetilde{\Psi}_V(k,t)e^{-i\hbar k^2\Delta t/2m},\tag{10}$$

where $\mathcal{F}(T)$ is the Fourier transform of T and $\widetilde{\Psi}=\mathcal{F}(\Psi)$.

For simplicity, we will now consider $\hbar=m=1$. As a final step, to obtain the evolution of the wavefunction after one timestep, we invert the Fourier transform from equation (10) and substitute for Ψ_V from equation (9). This leads to:

$$\Psi(x,t+\Delta t)=\mathcal{F}^{-1}\bigg(\exp\Big(-ik^2\Delta t\Big)\cdot\mathcal{F}\Big(\exp(-iV(x)\Delta t)\Psi(x,t)\Big)\bigg).$$

The accuracy of this method can be improved by splitting the action of the potential operator into two symmetric parts, as follows:

$$\Psi(x,t+\Delta t)=\exp\Big(-iV(x)\frac{\Delta t}{2}\Big)\mathcal{F}^{-1}\bigg(\exp\Big(-ik^2\Delta t\Big)\cdot\mathcal{F}\Big(\exp(-iV(x)\frac{\Delta t}{2})\Psi(x,t)\Big)\bigg).$$

This method is also known as Symmetrized Split-Step Fourier method [3].

3 Comparison with analytic solutions

3.1 Free particle

Finding the analytic solution for a Gaussian wavepacket with initial momentum $p_0 = \hbar k_0$ in zero potential ($V = 0$) involves determining the inverse Fourier Transform of a Gaussian function, centered on k_0 in k-space:

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \Phi(k)\, e^{i\left(kx - \omega(k)t\right)} dk,$$

where $\Phi(k)$ and $\omega(k)$ are given by:

$$\Phi(k) = N\, e^{-\frac{(k-k_0)^2}{2\sigma^2}} \quad \text{and} \quad \omega(k) = \frac{\hbar k^2}{2m}.$$

Completing the square and using the result for the standard Gaussian integral, we can obtain the general solution:

$$\big|\, \Psi(x,t)\, \big|^2 = N^2 \Big(\frac{1}{\sigma^2} + \frac{\hbar^2 t^2}{m^2}\Big)^{-1/2} \exp\Big(\frac{-4m^2\sigma^2x^2 - 4\hbar^2\sigma^2k_0^2t^2 + 8\hbar\sigma^2tmk_0x}{4m^2 + 4\hbar^2\sigma^4t^2}\Big), \tag{11}$$

which has an initial condition:

$$\Psi(x,0) = N\sigma\, e^{-\frac{\sigma^2x^2}{2}}. \tag{12}$$

In order for $\big|\, \Psi(x,0)\, \big|^2$ to be normalized, we calculate the necessary constant to be $N = (\pi\sigma^2)^{-1/4}$.

Plugging N into equation (11) and plotting both $\big|\, \Psi(x,t)\, \big|^2$ and our numerical solution, generated using the Backward Euler method for the initial condition given in (12), we obtained a difference between the analytic and numerical result of order 10^{-3}, which offers a satisfactory approximation. This is more clearly illustrated in Figure 1.

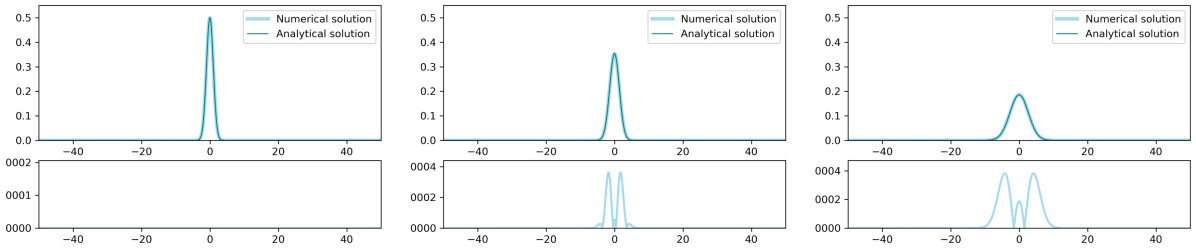


Figure 1: Snapshots showing the broadening of the probability distribution of an initial Gaussian wavepacket in zero potential (in the upper part) and the difference between the analytic and the numerical solution - generated using Backward Euler (in the lower part).

3.2 Infinite potential well

Choosing the walls of infinite potential to be on 0 and L, we know the stationary states of a particle inside this box and their corresponding energys are given by:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\Big(\frac{n\pi x}{L}\Big) \quad \text{and} \quad E_n = \frac{1}{2m}\Big(\frac{n\pi\hbar}{L}\Big)^2.$$

Due to the fact that these stationary states form a complete basis, any initial function $\psi(x)$ can be written as a superposition of different normal modes:

$$\psi(x) = \sum_{n=0}^{\infty} q_n \psi_n(x) \quad \text{and} \quad q_n = \sqrt{\frac{2}{L}} \int_0^L \psi(x) \sin\Big(\frac{n\pi x}{L}\Big) dx.$$

Being a superposition of stationary states of known coefficients, the time evolution of the wavepacket can now then be determined from:

$$\Psi(x,t) = \sum_{n=0}^{\infty} q_n \sqrt{\frac{2}{L}} \sin\Big(\frac{n\pi x}{L}\Big)\, e^{-i\frac{E_n}{\hbar}t}.$$

Given that this calculation is being done on a computer, a certain degree for n has been chosen instead of infinity ($n = 100$), after making sure it adequately approximated the initial function.

Using the Crank-Nicolson method to obtain the numerical solution for an infinite potential well problem, as illustrated in Figure 2, and comparing it with the analytic solution, we obtained a difference of order 10^{-3}, which indicates that the approximation is again satisfactory.

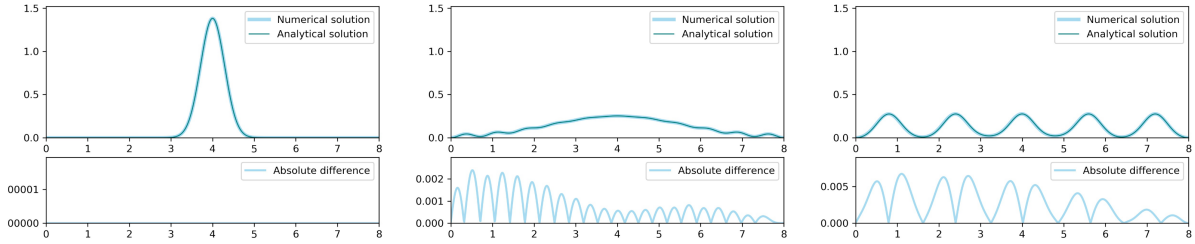


Figure 2: Snapshots showing the time evolution of the probability distribution of an initial Gaussian wavepacket in an infinite potential well (in the upper part) and the difference between the analytic and the numerical solution - generated using Crank-Nicolson (in the lower part).

3.3 Harmonic oscillator

Finding the analytic solution of the quantum harmonic oscillator follows a similar reasoning to what was previously presented in subsection 3.2, namely writing the initial wavefunction as a superposition of stationary states and then computing the evolution of the system by summing

over the time evolution of all the states considered.

The stationary states of a particle subject to a harmonic potential and the corresponding energy levels are given by:

$$\psi_n(x) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega x^2}{2\hbar}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) \quad \text{and} \quad E_n = \frac{\hbar\omega}{2}(2n+1),$$

where H_n stands for Hermite polynomial of order n.

Due to the orthogonality of these polynomials, the set of coefficients q_n corresponding to each stationary state $\psi_n(x)$ can be computed, after considering $\psi(x) = \sum_{n=0}^{\infty} q_n \psi_n(x)$. Therefore, the time evolution of any function $\Psi(x,0)$ is determined to be:

$$\Psi(x,t) = \sum_{n=0}^{\infty} \left(\int_{-\infty}^{\infty} \Psi(x,0)\, \psi_n(x)\, dx \right) \psi_n(x)\, e^{-i\frac{E_n}{\hbar}t}.$$

After choosing an adequate order for the number of terms in the sum ($n = 80$), we obtained the numerical solution for a wavepacket with initial momentum, using the Split Step Fourier method, as shown in Figure 3. The fact that the difference between this and the analytic solution is only of order 10^{-5} indicates a very precise approximation, performing better than the previously used methods.

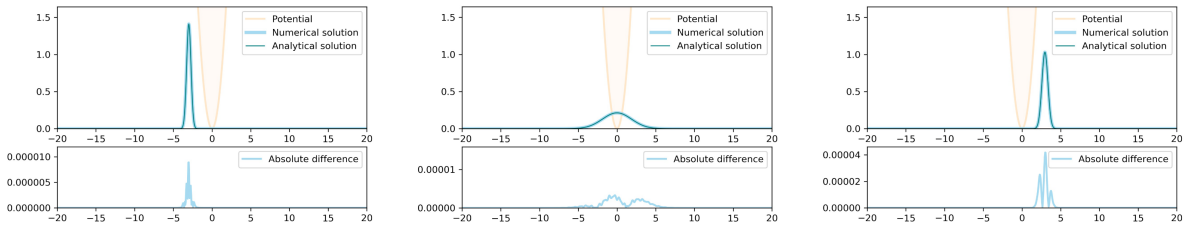


Figure 3: Snapshots showing the time evolution of the probability distribution of a Gaussian wavepacket with initial momentum in negative x-direction, interacting with a harmonic potential (in the upper part) and the difference between the analytic and the numerical solution - generated using Split Step Fourier (in the lower part).

4 Comparison of numerical methods

Starting with the Forward Euler method (2.1), it correctly approximated the analytic solution until the point where it starts to break because of the constrained ratio of the steps, which is given in equation (5). This led us to trying the implicit Euler method (2.2), which is not restricted by the steps used and is stable for longer intervals of time. A comparison between the

two methods can be seen in Figure 4. It shows snapshots of the two animations generated using Forward and Backward Euler, respectively, demonstrating their level of stability.

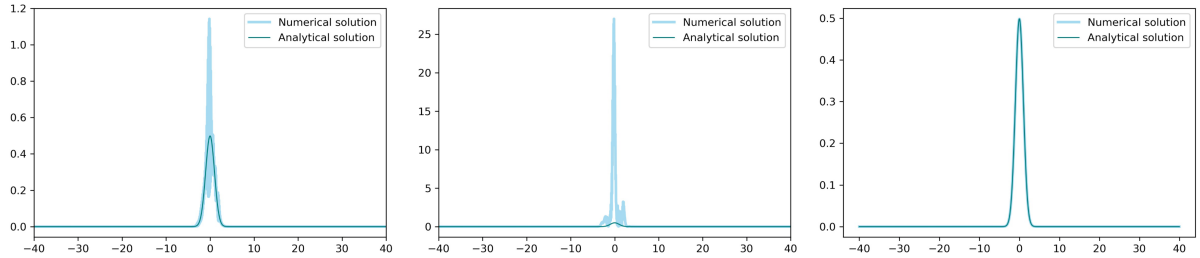


Figure 4: The first two plots are obtained from Forward Euler with steps $\Delta x = 0.05$ and $\Delta t = 0.001$ (at frames 50 and 60) and the third one is obtained from Backward Euler, with the same steps and initial condition, at frame 55.

As the Backward Euler method proved to follow the analytic solutions to known potentials (as seen in section 3), the case of a time-dependent potential was also attempted. However, the results provided by this method clearly indicated that the energy of the system was not conserved, and therefore the initial wavefunction would lose out in time (Figure 5).

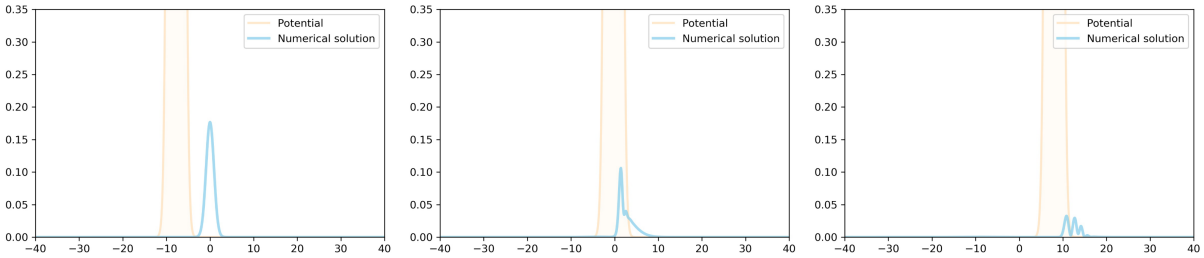


Figure 5: Snapshots of the animation generated by the Backward Euler method with steps $\Delta x = 0.05$ and $\Delta t = 0.05$, showing the probability distribution of an initial Gaussian wavepacket subject to a moving wall of potential, at frames 15, 65 and 120.

As the energy was greatly lost over time, we then moved to another numerical method, Crank-Nicolson (2.3). Combining both the Forward and the Backward Euler schemes, this method is very stable (due to the implicit part) and also more accurate (due to the explicit part). However, over long intervals of time, it still loses out, proving that it does not conserve the energy either. A comparison between Backward Euler and Crank-Nicolson is given in Figure 6, showing that after the same interval of time (measured in number of frames), the latter loses out less than the former.

This led us to trying a different numerical method, called Split Step Fourier (2.4), which avoids the discrete differentiation by switching to the Fourier k domain. This is a method of near-

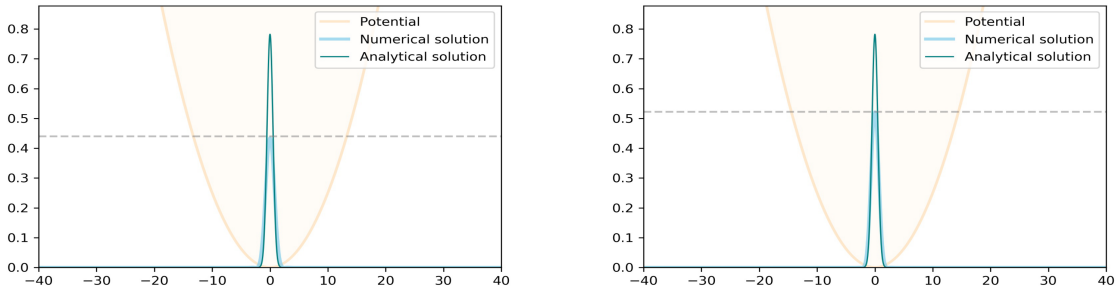


Figure 6: Two snapshots of the animations generated using Backward Euler (on the left) and Crank-Nicolson (on the right) methods, for a Gaussian wavepacket in a harmonic potential, both with steps $\Delta x = 0.05$ and $\Delta t = 0.05$. It can be seen that at the same frame, for an initial Gaussian peaked at 0.8, the former reaches 0.44, whilst the latter reaches 0.52.

conservation of energy on long time intervals and therefore it allowed probing time dependent potentials, as can be seen in the following figure:

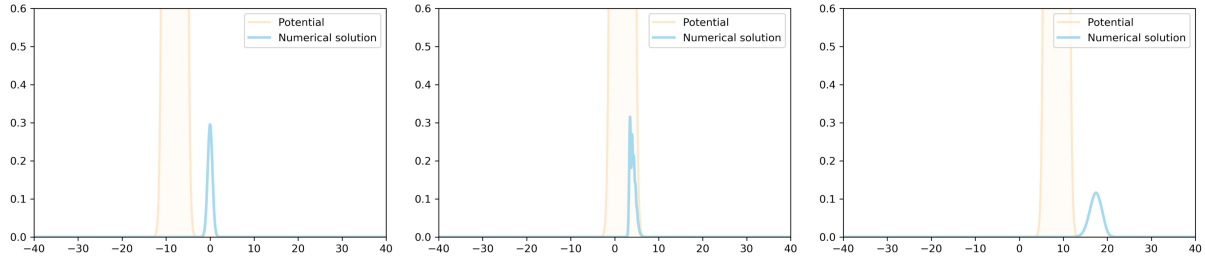


Figure 7: Snapshots of the animation generated by the Split Step Fourier method with steps $\Delta x = 0.05$ and $\Delta t = 0.01$, showing the probability distribution of an initial Gaussian wavepacket being reflected off a moving wall of potential, at frames 20, 118 and 185.

In order to quantify the difference between the analytic and each numerical solution, we used two different methods [4]. The one used in measuring the similarity for a Gaussian wavepacket in a harmonic potential implied the calculation of the area between the two curves and plotting its evolution in time for each numerical method used, as illustrated on the right side of Figure 8. As for the evolution of a free wavepacket, calculating the difference in curve lengths proved to be more efficient, the results obtained being shown on the left side of Figure 8.

From these plots, it can easily be inferred that the accuracy of Backward Euler and Crank-Nicolson methods decreases with time. Also, the Forward Euler method proves to be the most unreliable as it diverges very rapidly. The Split Step Fourier numerical method turns out to be the most precise and stable, as the difference between it and the analytic solution is very close to 0, even for long intervals of time.

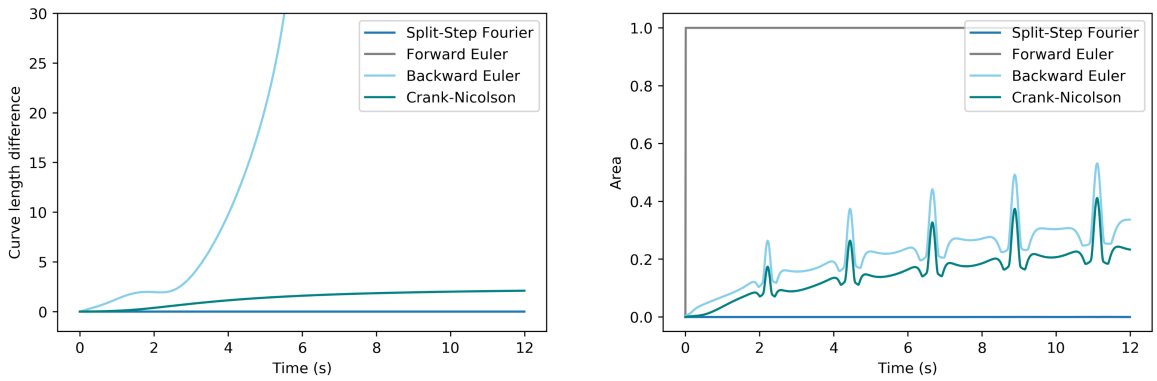


Figure 8: Comparison between the numerical and analytic solutions, quantified by the curve length difference (on the left) for a Gaussian wavepacket in zero potential, and by the area between the curves (on the right) for the same wavepacket, but in a harmonic potential.

5 Other potentials

5.1 Morse potential

The evolution of a Gaussian wavepacket was studied in the potential of a diatomic molecule, given by the Morse potential. This potential offers a better approximation for the interatomic interaction than the quantum harmonic oscillator, as it accounts for the dissociation energy and anharmonicity [5]. In mathematical terms, the potential has the following form:

$$V(x) = D_e \left(1 - e^{-a(x-x_e)}\right)^2,$$

where D_e is the well depth, a determines the width and x_e is the equilibrium position. It can be seen from Figure 9 that the symmetry of the initial wavepacket is lost, due to the difference between the shape of the Morse potential and the harmonic one.

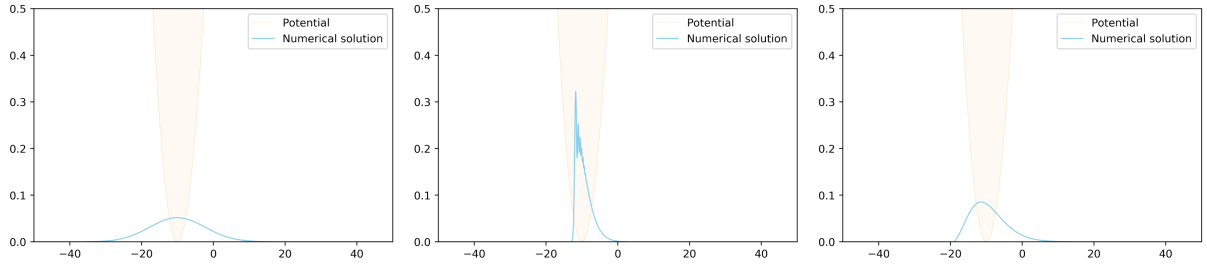


Figure 9: Snapshots showing the time evolution of the probability distribution of an initial Gaussian wavepacket in the Morse potential, generated using the Split Step Fourier method.

5.2 Finite barrier - tunnelling

One of the most intriguing phenomena arising from quantum theory is quantum tunneling. It refers to the process in which a particle passes through a high enough potential barrier, that classically described would not be allowed. Tunnelling is essential in several areas of physics, such as quantum biology, where it is a key factor in cellular respiration and photosynthesis, quantum computing, nuclear fusion in stars and many others [6].

Using the Fourier Split Step method, we analysed the behaviour of a Gaussian wavepacket directed at a potential barrier. The behaviour of such a wavepacket is shown in Figure 10, where the tunnelling phenomenon can be observed, as the probability on the other side of the barrier is different than 0 (on the right side of the figure).

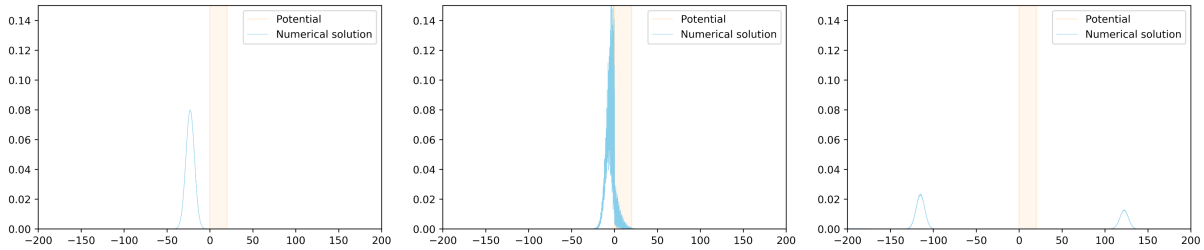


Figure 10: Snapshots representing the probability distribution from the simulation of an incident Gaussian wavepacket on a high potential barrier, generated using the Split Step Fourier method.

5.3 Periodic potentials

One problem of high interest in solid state physics is the nearly free electron model, which assumes a weak interaction between a travelling electron through a periodic crystal lattice. For our one-dimensional problem, we modelled the potential as $V(x) = A\sin(\omega x)$ and we simulated the time evolution of a travelling Gaussian wavepacket interacting with it (Figure 11).

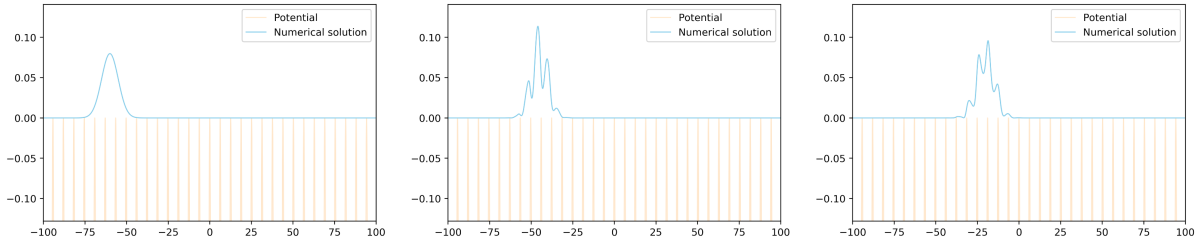


Figure 11: Snapshots showing the time evolution of the probability distribution of a Gaussian wavepacket travelling through a periodic potential, generated using Split Step Fourier.

5.4 Time dependent potentials

As the method of separation of variables ceases to solve the Schrödinger equation for a time dependent potential, the most convenient approach that can be used to obtain a solution is by numerical methods. Figure 12 illustrates the behaviour of an initial Gaussian wavepacket in a time dependent potential of the form:

$$V(x,t) = \frac{1}{2}x^2 + A\sin(\omega t)e^{-8x^2}.$$

Similar potentials, often called "Mexican hat" potentials, are studied in quantum field theory in modelling spontaneous symmetry breaking [7].

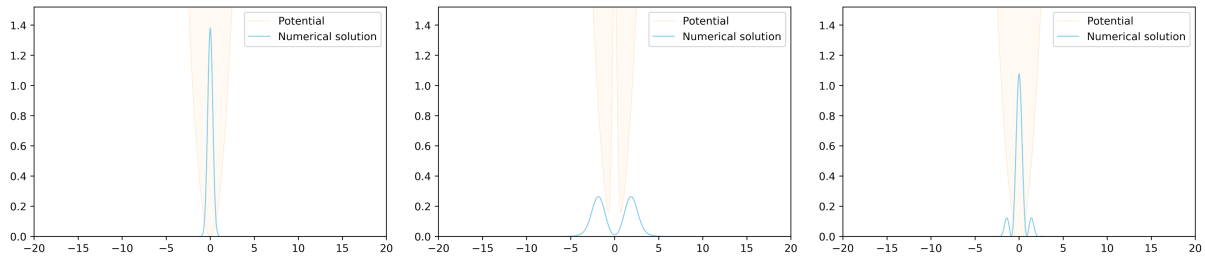


Figure 12: Snapshots illustrating the time evolution of the probability distribution of a Gaussian wavepacket in an oscillating time dependent potential, generated using Split Step Fourier.

6 Conclusions

In this project, we implemented four different numerical methods in order to obtain solutions to the one-dimensional Schrödinger equation. Comparing those with the analytic solutions, where possible, we were able to obtain valuable information about the stability and efficiency of the numerical methods used. We came to the conclusion that the Split Step Fourier method is the most efficient in terms of both stability and accuracy. The Crank-Nicolson and Backward Euler methods offer a satisfactory approximation for short enough intervals of time, while the Forward Euler method proved to be unreliable and very unstable. As for future improvements, extending the applications in two dimensions can produce more relevant results, as well as modelling the phase space of the dynamical systems considered.

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